

Keeping Bias Out of Lending AI

What it actually takes, in consumer credit and mortgage origination

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1 THE QUESTION

Lenders are deploying models into two decisions that shape household wealth: **consumer installment credit** and **residential mortgage origination**. Both are regulated — ECOA and Regulation B govern both, the Fair Housing Act adds to mortgage. The question is not whether bias matters here but what a team must actually *do* to keep it out, and whether the standard toolchain gets them there.

We built a full audit for both sectors on real data (1,000 consumer credit and 10,978 Georgia mortgage applications) and measured what it took to make that audit correct. Most of what we learned came from the ways it was wrong first.

2 WHAT THE STANDARD TOOLCHAIN DOES AND DOES NOT GIVE YOU

We checked AIF360 0.6 and Fairlearn 0.13 directly rather than assuming.

Which outcome is “favorable”? AIF360 defaults to `favorable_label=1, 0`. On a target named `loan_default` that means *defaulting is the good outcome*. We reproduce it: the default returns $DI = 1.2072$ where the correct orientation returns 0.9862. Fairlearn has no favorable-label parameter at all. Credit-risk targets are almost always adverse-coded, so the default is almost always wrong.

How much data per subgroup? Neither toolkit has *any* concept of minimum subgroup size — no floor, no warning, no guidance. Every team invents its own.

Who counts as protected? Neither binds a protected class to a statute. You pass a column; nothing checks it against ECOA. Nor does either record which reference group produced a ratio.

The toolkits compute correctly. They do not stop you answering four regulatory questions by accident.

3 WHAT WE GOT WRONG, AND WHAT IT COST

Because nothing supplied them, we invented our own floor ($n \geq 15$), severity bands, and output checks. Each changed a conclusion.

Sample size inverted a disparity. Auditing a 2,196-row sample, our mortgage audit ranked **American Indian men as the best-treated group** ($DI = 1.1504$) — on four applicants who all happened to be approved. Over all 10,978 they are the **worst** group in the data (0.6734). The floor was not the error: the cell it suppressed, at 0.1917, was itself noise (0.8815 at full population). At small n the estimates are wrong in *both* directions, and no floor can tell the cases apart.

We audited age under the wrong statute. We used 40+ — that is the ADEA, employment law. Credit is ECOA, and Regulation B §1002.2(o) protects 62+. On our threshold the model looked clean (0.9791); on the statutory one it shows the largest age disparity in

the data (0.8897). The flag we needed was in the raw HMDA file, unused.

Our automated checks punished correct answers. A 40-character minimum marked 6 correct answers as refusals, then zeroed 14 evaluations of a second AI provider ($70.0 \rightarrow 19.79$) — on which we blocked that vendor. A fabrication check flagged a newer model 5× more often, but 67% of those flags were arithmetic it was asked to do (88.88% is 0.8888).

4 WHAT IT TAKES: A PROTOCOL FOR THESE TWO SECTORS

1. State the favorable outcome explicitly, every time; never accept the default. **2. Audit the book, not a test split** — in mortgage, HMDA already requires you to file it. **3. Bind the protected class to the governing statute in code**, citation beside the threshold: ECOA 62+, not ADEA 40+. Note the asymmetry a parity metric cannot express — §1002.6(b)(2) permits *favouring* elderly applicants, so a ratio above 1 is lawful. **4. Publish what you suppressed**, with n and the unreported ratio; a silent floor is indistinguishable from a clean result. **5. Record the reference group** beside every ratio — most-favoured-cell versus designated control moves verdicts across the four-fifths line. **6. Test every automated gate where the right answer is unusual**: the terse answer, the small subgroup, the model that shows its work. All three of ours failed exactly there.

Why it matters. Every failure above produced a *clean* result — the dangerous direction in lending. A false alarm costs a reviewer an afternoon; a false clearance closes the question and the applicant never learns why. An audit that cannot say what sample it ran on and what it withheld cannot support the clearance it is used to give.

5 ARTIFACTS

`artifacts/consolidated/RESULTS.md · report/report_facct.tex · scripts/compare_populations.py`. Regenerates offline, no API key, \$0.89 of compute; 60 tests green.